Semester Project Documentation

Semester Project Title: Virtual Pet Simulator

Git Link: https://github.com/izzzaaaaa/Virtual-Pet-Simulator.git

Student Details

	Student Name	Student Reg #	Student Degree
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Main Features

1) Feed the pet
2) Play with the pet
3) Put pet to bed
4) Multiple pet types
5) Game manager
6) Save and load game progress

Types of Users

Students (Educational Context), Developers

Requirements Breakdown

1.1) Give the pet a certain amount of food to keep it alive and happy.

2.1) Play with the pet to keep it happy.

3.1) Make the pet sleep to regenerate energy.

4.1) Choose from multiple pets:

- *4.1.1)* Cat
- *4.1.2)* Dog
- *4.1.3) Dragon*

5.1) Game manager to run the game.

6.1) File handling to save and load game progress.

Features to Codding Matrix

Sr no.	Feature Name	Concept Used	Functions Created	Variables / Obj Created	Line of Code Written
1	Pet	Base Class	makeSound() draw() getType() feed(float amount) play() sleep() update(float deltaTime) isAlive() isHungry() isTired() isSick() getName() getHunger() getEnergy() getHappiness() getHealth() getAge() setName(string n) setHunger(float h) setEnergy(float e) setHappiness(float h) setHealth(float H) setAge(int a) operator<<(ostream& os, const Pet& pet)	string name float hunger float energy float happiness float health int age bool alive	186
2	Cat	Derived Class	purr() scratch()	float playfulness	57
3	Dog	Derived Class	bark() fetch()	float loyalty	58
4	Dragon	Derived Class	breatheFire()	float fireLevel	50
5	Game Manager	Base Class	run() handleInput() update() displayStats() showControls() createPetMenu() usePetAction(char choice) saveGame() loadGame()	Pet* activePet bool isRunning string petName	326

6	SaveSystem	Base Class	saveGame(const Pet& pet) loadGame(Pet& pet)	ofstream file ifstream file	50
7	main		main()	GameManager game	8